

# Question Paper

Exam Date & Time: 22-Dec-2025 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026

SUBJECT: ECE 2127/EC\_ 2127 DIGITAL CIRCUITS

(SPL: ELECTRONICS ENGINEERING (VLSI DESIGN AND TECHNOLOGY))

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Design a combinational circuit using 4:16 decoder with active-high outputs to convert 8421 BCD code to equivalent 2421 BCD code. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 1B) Implement the following two functions:  $F_1 = \sum m(1, 2, 4, 7)$  and  $F_2 = \sum m(3, 5, 6, 7)$  using a programmable logic array (PLA). (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 1C) Design a minimal cost product-of-sums expression for the function  $f(a, b, c, d) = \sum m(0, 2, 4, 5, 6, 7, 8, 10, 12, 14, 15)$  using Boolean algebra. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Design a Mealy-type circuit, using D flip-flops, with one input and one output, to detect the overlapping occurrences of the bit pattern 1010 in a sequential pulse train. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 2B) Implement the Boolean function  $F = A\bar{B} + BD + \bar{B}C\bar{D}$  using 4:1 multiplexer by connecting the inputs A & B to the select lines of the MUX. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) Using the characteristic equations and excitation table, convert a D flip-flop into a JK flip-flop. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 3A) Derive the boolean expressions for the next state logic of a hazard-free mod-4 asynchronous pulse counter. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) Using negative edge triggered JK flipflops, design a decade ripple up-counter and draw its timing diagram. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 3C) Construct the excitation table for a Master-Slave D flip-flop as an asynchronous sequential circuit using the Moore approach. Use present state variables 'ym' for the master and 'ys' for the slave, with 'Clock' and 'D' as inputs. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) Design a synchronous Mod-8 down counter using JK flipflops. Draw the timing diagram for the same. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4B) Using a universal shift register and logic gates, design a circuit to generate the following sequence: 1001110100111010... (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4C) Design a sequential circuit to generate the following sequence 100111100111100.. using indirect logic. (3)

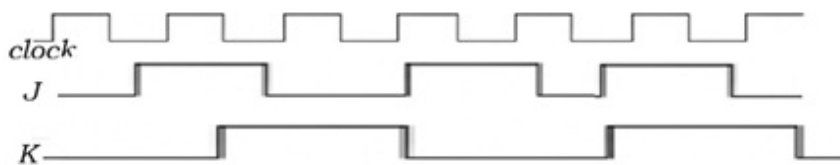
Course Outcome: "CO5" / Cognitive Level: "Application"

- 5A) Design an optimal synchronous sequential circuit based on the given state table using D flip-flops: (4)

| Present state | Next state |         | Output $z$ |
|---------------|------------|---------|------------|
|               | $w = 0$    | $w = 1$ |            |
| A             | B          | C       | 1          |
| B             | D          | F       | 1          |
| C             | F          | E       | 0          |
| D             | B          | G       | 1          |
| E             | F          | C       | 0          |
| F             | E          | D       | 0          |
| G             | F          | G       | 0          |

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Plot the timing diagram showing the Q and  $\bar{Q}$  outputs of a level triggered JK flipflop, with the control input clock, based on the given waveform. (3)



Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Design a synchronous sequential circuit that generates the following sequence of numbers:  $1 \rightarrow 4 \rightarrow 3 \rightarrow 5 \rightarrow 2 \rightarrow 6 \rightarrow 1 \rightarrow \dots$  using edge-triggered D flip-flops. Assume the circuit does not enter any undesired states. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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| Event Name | Subject |
|------------|---------|
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECE\_2127:Digital Circuits

| #     | Course Outcome | Description | Marks |
|-------|----------------|-------------|-------|
| 1     | 1              | CO 1        | 3     |
| 2     | 3              | CO 3        | 10    |
| 3     | 4              | CO 4        | 15    |
| 4     | 5              | CO 5        | 22    |
| Total |                |             | 50    |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
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058-MIT BLR-MAKEUP EXAM-2025-  
2026

ECE\_2127:Digital Circuits

| # | Level       | No of Questions | Marks (%) |
|---|-------------|-----------------|-----------|
| 1 | Remember    | 0               | 0(0%)     |
| 2 | Understand  | 0               | 0(0%)     |
| 3 | Application | 15              | 50(100%)  |
| 4 | Analysis    | 0               | 0(0%)     |
| 5 | Create      | 0               | 0(0%)     |
| 6 | Evaluate    | 0               | 0(0%)     |
|   | Total       | 15              | 50 (100%) |